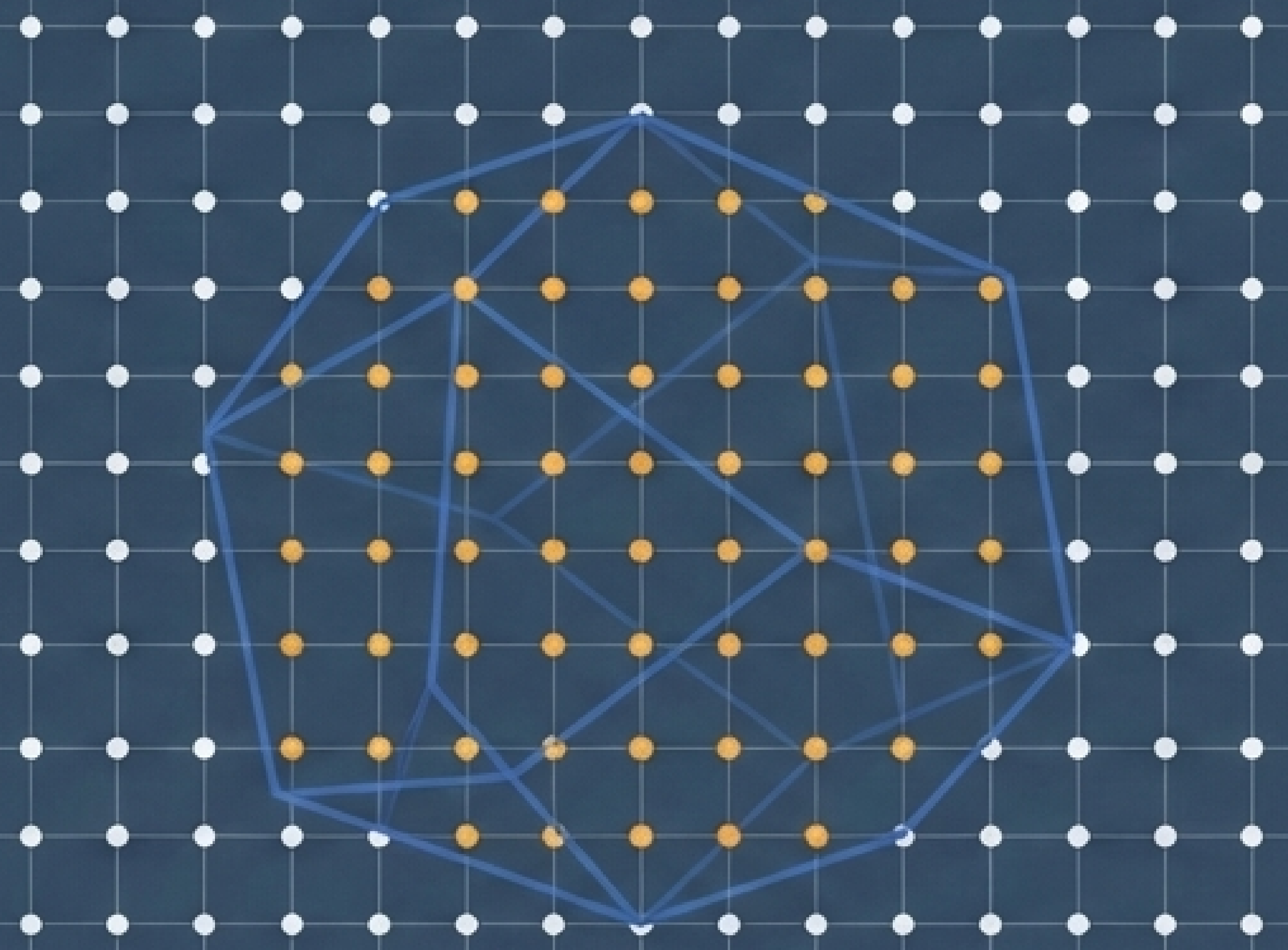


THE ARCHITECTURE OF OPTIMIZATION

FORMULATING LINEAR, INTEGER, AND MIXED-INTEGER PROGRAMMING MODELS

MATHEMATICAL OPTIMIZATION MODELING

Presentation Outline

1. Linear Programming Basics

2. Integer Variables & Logical Constraints

3. modeling techniques

4. MILP

NotebookLM

Defining Linear and Non-Linear Programming

Linear Programming (LP)

Objective function and all constraints are linear functions of the decision variables.

Example (e.g.):

$$\max Z = 2x + 3y$$
$$\text{s.t. } x + y \leq 5$$
$$x, y \geq 0$$

Non-Linear Programming (NLP)

At least one of the objective function or constraints is a non-linear function.

Example (e.g.):

$$\max Z = 2x^2 + 3y$$
$$\text{s.t. } x^2 + y \leq 5$$
$$x, y \geq 0$$

Linear Programming (LP) Fundamentals

Decision Variables

$x \in \mathbb{R}^n$

Objective Function

$\min/\max \ c^T x$

Constraints

$Ax \leq b, Ax = b, Ax \geq b$

The Convex Guarantee

Because the feasible region is a convex polyhedron, any local optimum found by Simplex or Interior-Point algorithms is mathematically guaranteed to be a global optimum. Auxiliary variables are introduced to transform nonlinear expressions into this linear framework.

Example: Factory Production

[Problem]

A factory produces three products (P1, P2, P3). We must maximize profit under strict limits: 150 labor hours and 200 material units.

[Components]

Decision Variables:

$$x_1, x_2, x_3 \geq 0$$

(Continuous production quantities)

Profit/Unit: \$40 (P1), \$30 (P2), \$50 (P3)
Labor/Unit: 2 (P1), 1 (P2), 3 (P3)
Material/Unit: 3 (P1), 2 (P2), 4 (P3)

[LP Formulation]

Objective:

$$\max\ z = 40x_1 + 30x_2 + 50x_3$$

Subject To (s.t.):

$$2x_1 + 1x_2 + 3x_3 \leq 150$$

(Labor Limit Constraint)

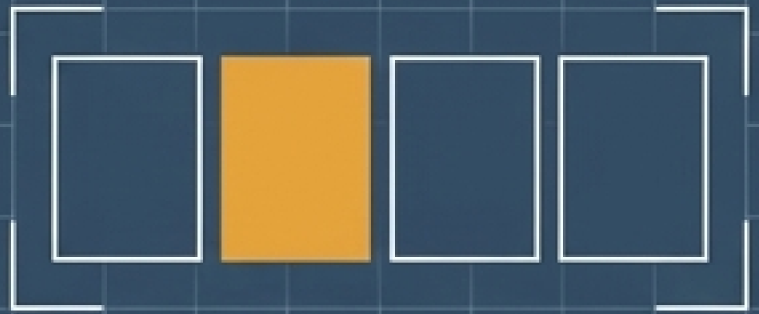
$$3x_1 + 2x_2 + 4x_3 \leq 200$$

(Material Limit Constraint)

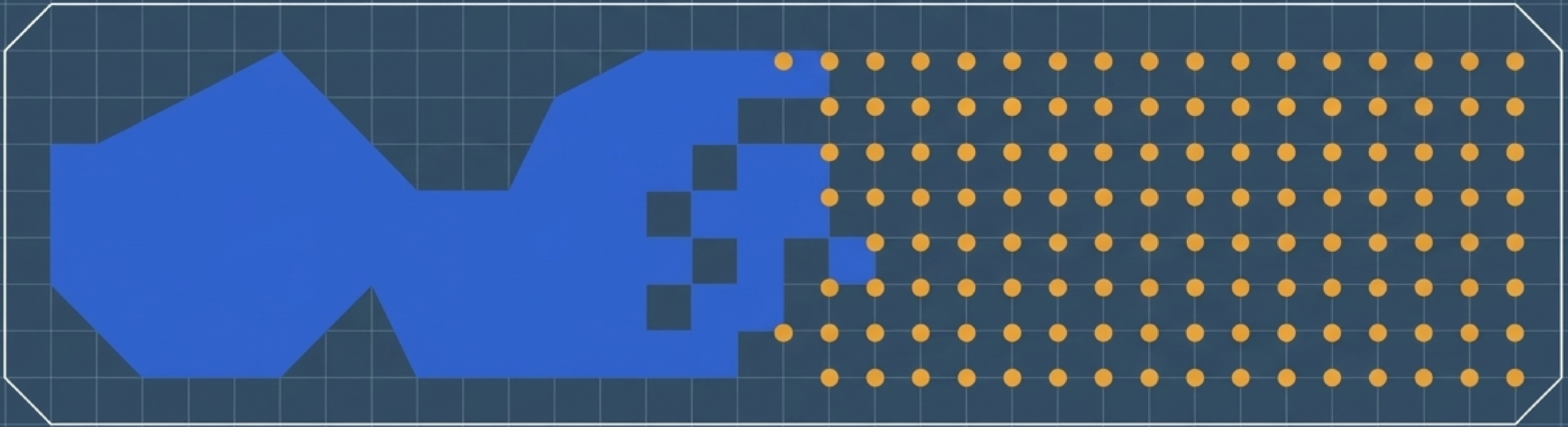
$$x_1, x_2, x_3 \geq 0$$

(Non-negativity)

Act II: The Discrete Complication



Continuous variables cannot represent binary choices like “yes/no” or “open/close”. We must introduce discrete integer models to capture real-world logical conditions, despite the computational cost.



	Linear Programming (LP)	Integer Programming (IP)	Mixed-Integer Linear Programming (MILP)
Variable Domains	Continuous $\mathbb{R}^n$	Discrete $\mathbb{Z}$ or Binary $\{0,1\}$	Mix of Continuous $\mathbb{R}^n$ and Discrete $\mathbb{Z}$ or Binary $\{0,1\}$
Feasible Region	Continuous convex polyhedron	Discrete, fragmented set of points	Union of disjoint polyhedra
Complexity	Solvable in polynomial time	Combinatorial and generally NP-hard	Combinatorial and generally NP-hard
Solvers	Simplex / Interior-Point	Branch-and-Bound / Cutting Planes	Branch-and-Cut / Branch-and-Bound

Translating Logic to Linear Inequalities

	Logical Concept	Boolean Logic	Linear Constraint
Category 1: Mutually Exclusive / Inclusion			
1	At least one is true	$x_1 \vee x_2 = 1$	$x_1 + x_2 \geq 1$
2	Exactly one is true	$(x_1=1, x_2=0) \vee (x_1=0, x_2=1)$	$x_1 + x_2 = 1$
3	Both must be true	$x_1 \wedge x_2 = 1$	$x_1 = 1, x_2 = 1$
Category 2: Implication (If-Then)			
4	If $x_1=0$, then $x_2=1$	$x_1=0 \Rightarrow x_2=1$	$x_2 \geq 1 - x_1$
5	If $x_1=0$, then at least two of (x_2, x_3) are 1	$x_1=0 \Rightarrow x_2+x_3\geq 2$	$x_2 + x_3 \geq 2(1 - x_1)$
Category 3: Cardinality Bounds			
6	At most 'r' variables are 1	-	$\sum x_i \leq r$
7	If $x_1=0$, at least 'r' variables are 1	-	$\sum x_i \geq r(1 - x_1)$

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Example: Store Opening with Logical Constraints

[Problem]

Evaluate 6 locations (L1-L6). Open a maximum of 4 stores. Minimize fixed and staffing costs.

[Logical Translation]

Rule: Max 4 stores → $\sum y_i \leq 4$

Rule: L1 and L2 cannot both open → $y_1 + y_2 \leq 1$

Rule: L3 and L4 must open together → $y_3 - y_4 = 0$

[Components]

Binary Variables (Amber):

$y_i \in \{0,1\}$ (1 if Store i opens, 0 otherwise).

Continuous Variables (Cobalt):

$s_i \in \mathbb{Z}_{\geq 0}$ (Number of staff at Store i).

[Formulation]

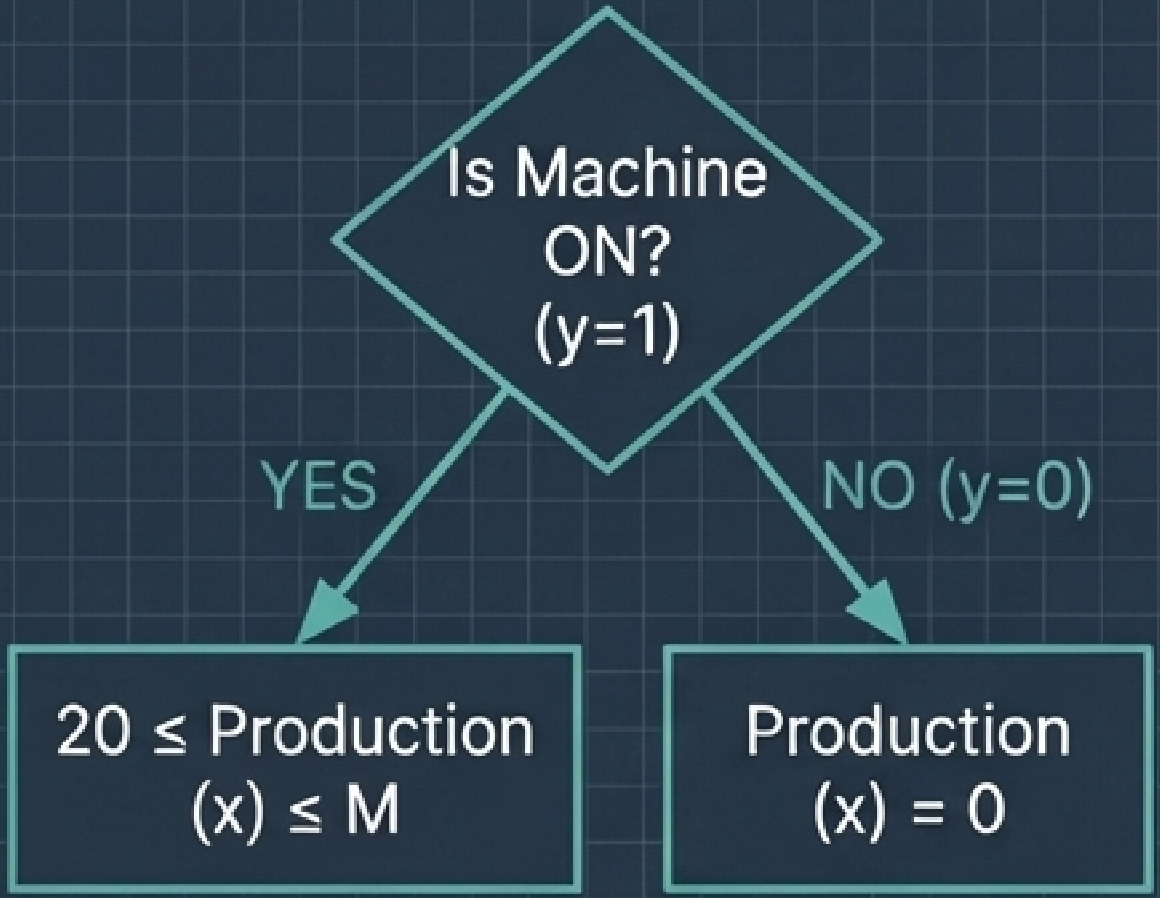
Objective Function (Z): Total Cost
Z = (Fixed Cost of Opening Stores) + (Variable Cost of Staffing)
$\min z = \sum(F_i y_i + c_i s_i)$
Goal: Minimize total cost Z by choosing stores (y_i) and staffing levels (s_i), subject to constraints.
Staffing Bounds: $3y_i \leq s_i \leq 10y_i$
(Staff must be between 3 and 10 only if store is open).

Act III: Transforming (MILP)

- **LP (Linear Programming):** Problems with continuous variables and linear constraints.
- **IP (Integer Programming):** Problems with integer variables and linear constraints.
- **MILP (Mixed-Integer Linear Programming):** Problems with a mix of continuous and integer variables.
- **Modeling Techniques:** Methods to translate logical rules and nonlinear costs into linear constraints for MILP solvers.

The Big-M Method: Activating Constraints Conditionally

Machine has M max capacity, 20 min capacity, M fixed cost, 5/unit variable cost.




```
if y == 1 then
   20 <= x <= M
else
   x == 0
```

Mathematical Formulation:

$$x \le M \cdot y$$
$$x \ge 20y$$
$$\min z = M \cdot y + 5x$$

Note: The large constant M forces x=0 when y=0, and relaxes the constraint when y=1.

Linearizing Min/Max & Piecewise Functions

Linearizing MIN/MAX

Simple Explanation: Linearization converts non-linear functions (like min/max) into linear equations that MILP solvers can understand.

Context: MILP solvers strictly prohibit min() or max() operators.

Defining z for z = min(x_A, x_B) (Upper Bounds): We first establish that z is less than or equal to both x_A and x_B.

$$z \leq x_A$$
$$z \leq x_B$$

Defining z for z = min(x_A, x_B) (Exact Extraction): To ensure z is exactly the minimum, we use a binary variable y and a large constant M. These constraints, combined with the upper bounds, force z to equal the smaller of x_A or x_B.

$$z \geq x_A - M(1-y)$$
$$z \geq x_B - My$$

Piecewise Functions (Project Bonus)

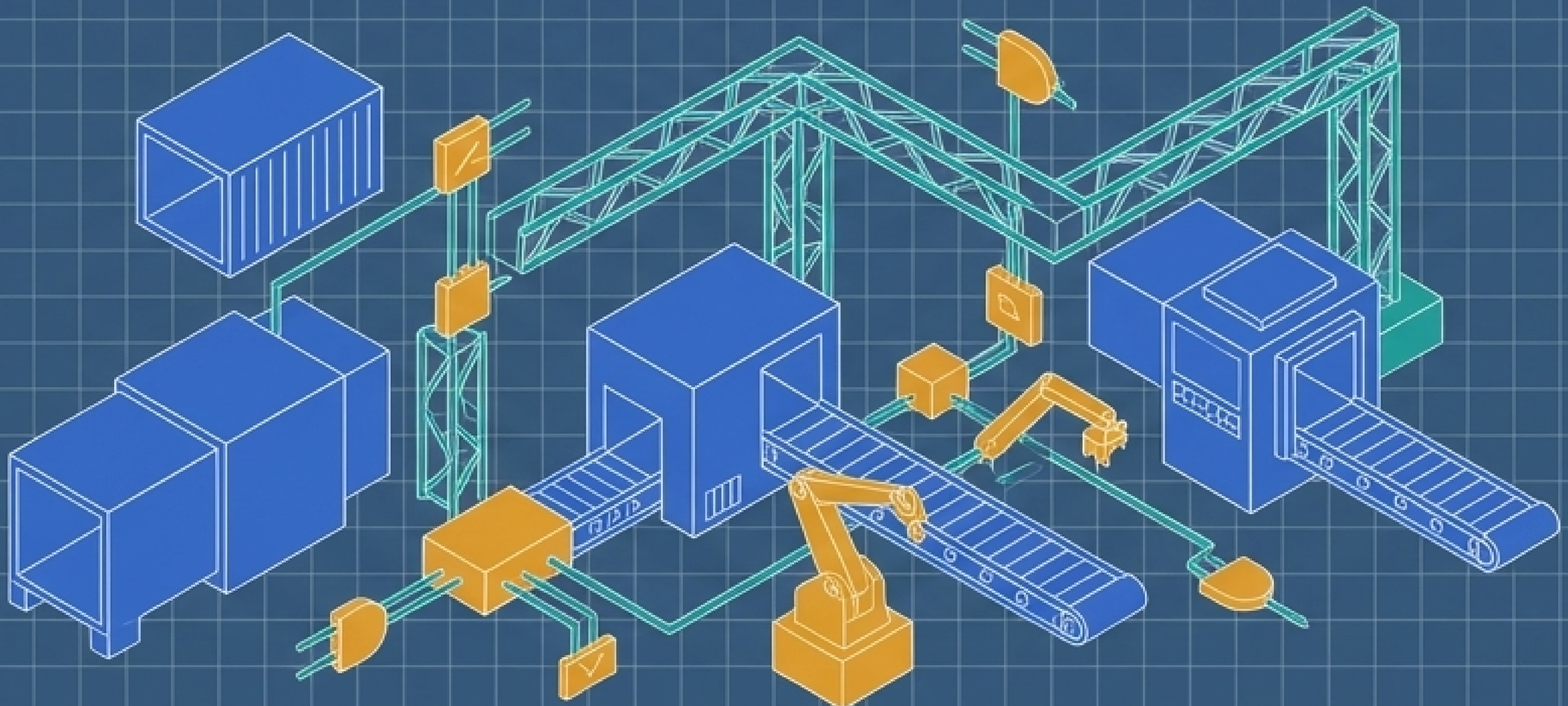
Context: Project A grants a 10-unit bonus only if budget x_A exceeds 50.

Formulation: Introduce binary indicator $y_A \in \{0,1\}$ for $x_A > 50$.

$$x_A \leq 50 + M_A y_A$$
$$x_A \geq 50 y_A$$
$$bonus_A = 10 y_A$$

Act IV: The Capstone

Having mastered continuous constraints, binary logic, and Big-M techniques, we now integrate them into a unified Mixed-Integer Linear Programming capstone model.



Comprehensive Example: MILP Formulation

[Problem]

3 products (P1, P2, P3), 2 machines (M1, M2).
Products require exclusive machine assignment.
Machines have conditional setup times.
Maximize net profit.

[Components] (Variables)

$y_j \in \{0,1\}$: Machine j ON/OFF state (Amber).

$a_{ij} \in \{0,1\}$: Product i assigned to Machine j (Amber).

$x_{ij} \geq 0$: Units of i produced on j (Cobalt).

[Formulation: Constraints]

Unique Assignment: $\sum_j a_{ij} = 1$

Conditional Logic: $a_{ij} \leq y_j$ (Assignment requires machine ON).

Big-M Capacity: $q_{min_i} a_{ij} \leq x_{ij} \leq q_{max_i} a_{ij}$ (Production bounded by assignment).

Total Labor: $10(y_1 + y_2) + \sum\sum \ell_i x_{ij} \leq 200$ (Setup + Variable).

[Formulation: Objective]

$\max z = \sum\sum p_i x_{ij} - \sum F_j y_j$

(Net Profit = Revenue - Setup Costs).

The Architecture Realized

Pillar 1
The Continuous Foundation (LP)

Linear formulations are not merely restrictions; they are mathematical guarantees. Convexity ensures that local discoveries are global truths.

Pillar 2
The Discrete Complication (IP)

Binary variables provide the descriptive power to encode the nonlinear reality of human decisions—turning continuous flows into rigid, logical choices.

Pillar 3
The Synthesizing Bridge (MILP)

Techniques like Big-M are the structural engineering of optimization, forcing **conditional reality into linear solvers to solve complex, integrated systems.**

Mastery of mathematical optimization is the ability to architect seamless bridges between the continuous laws of physics and the discrete logic of business.